

Mapping the Space Charge at Nanoscale in Dielectric Polymer Nanocomposites

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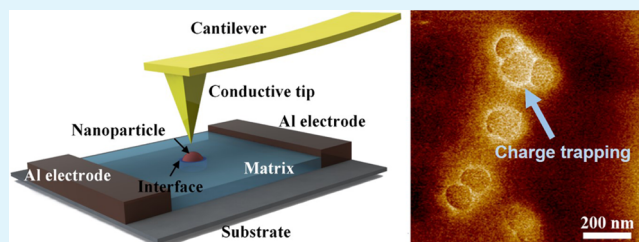
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ABSTRACT: Heterogeneous dielectric materials such as dielectric polymer nanocomposites have attracted extensive attention because of their exceptional insulating and dielectric performance, which originates from the unique space charge dynamics associated with the various interfacial regions. However, the space charge distribution and transport in polymer nanocomposites remain elusive due to the lack of analytical methods that can precisely probe the charge profile at the nanoscale resolution. Although a few studies have explored the possibility of using scanning probe techniques for characterizing the local charge distribution, the interference from induced electrical polarization of the material has been unfortunately ignored, leading to inaccurate results. In this contribution, we report an open-loop Kelvin probe force microscopy (KPFM) method with nanoscale resolution for the direct detection of the space charge profile in dielectric polymer nanocomposites. Unlike the conventional studies where a vertical direct current (DC) voltage is applied on the sample through the probe to evoke the charge injection and transport in dielectric polymer nanocomposites, the present method is established based on a delicate electrode configuration where a lateral electric field is allowed to be applied on the sample during the KPFM test. This special testing configuration enables real-time charge injection and transport without inducing the electrical polarization of material along the vertical direction, which gives rise to clean mapping of space charges and reveals the interfacial charge trapping in polymer nanocomposites. This work provides a robust and reliable method for studying the sophisticated charge transport properties associated with the various interfacial regions in heterogeneous dielectric materials.

KEYWORDS: nanocomposite, interface, dielectric, charge trapping, Kelvin probe force microscopy



1. INTRODUCTION

Dielectric polymers are ubiquitously utilized as electrically insulating materials and dielectric components in modern electronics and electrical systems. Space charge distribution and transport in dielectric polymers are critical factors that determine the insulating and dielectric performance.¹ Capitalizing on the well-recognized ability of inorganic nanoparticles to modulate the transport, adsorption, and trapping characteristics of charge carriers, the polymer nanocomposite approach has been widely practiced for promoting the various electrical performance metrics, such as breakdown strength, resistance to degradation, and electrical resistivity.^{2,3} Many studies have concluded that the performance improvement in dielectric polymer nanocomposites stems mainly from the charge trapping sites associated with the interfacial region,^{2,4–6} namely, a nanoscale transitional volume between the nanoparticles and the polymer matrix. The charge trapping sites in solid dielectric materials refer to the localized states in the forbidden band gap of their electronic structures, which are capable of capturing free charge carriers in the materials. Since the traps are believed to be related to the local structure of materials (e.g., physical and chemical defects, and impur-

ities),^{7,8} theories and models regarding the interfacial region have been proposed to explain the abnormal behavior of charge carriers in the nanocomposites. However, it remains elusive that how the charge transport and accumulation are affected by the embedded nanoparticles and the interfacial region in dielectric polymer nanocomposites.

To date, the understanding of interfacial charge behavior in polymer nanocomposites has largely relied on theories and indirect experimental results, principally because the nanoscale interface is beyond the spatial resolution of traditional analytical techniques. To be specific, the currently employed methods for characterizing charge carrier traps either lack the ability to spatially resolve the trapped charges, such as photon-stimulated current spectroscopy (PSC)⁹ and thermally stimulated depolarization current (TSDC),¹⁰ or the spatial

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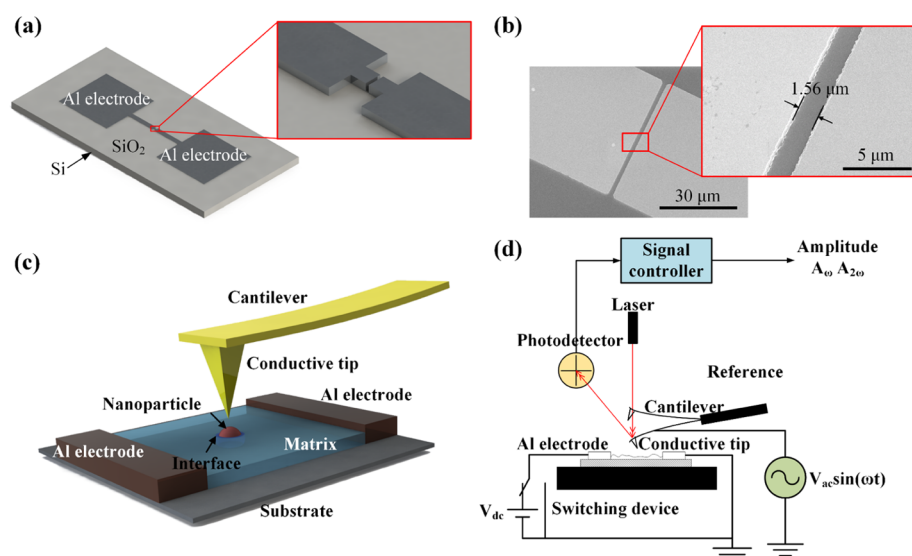


Figure 1. (a) Schematic diagram of the substrate. (b) SEM images of the narrow slit between the two Al electrodes. (c) Schematic diagram showing the nanocomposite film between the electrodes. (d) Schematic diagram of the KPFM testing configuration.

resolution is limited to above tens of micrometers, including isothermal surface potential decay (ISPD),¹¹ early Kelvin probe methods,^{12,13} laser-induced pressure pulse (LIPP), pulsed electroacoustic (PEA) methods, etc.^{14,15} And such limitation is usually inherent to the working principle of the analytical techniques, e.g., the spatial resolution of PEA measurement is related to the velocity of the acoustic wave.

The established methodologies based on atomic force microscopy (AFM) have enabled nanoscale mapping of a multitude of material properties.¹⁶ Electrostatic force microscopy (EFM) and Kelvin probe force microscopy (KPFM) are particularly powerful for the detection of local electrical properties through the measurement of electrostatic forces between the AFM tip and the sample.^{17–23} While the detection of electric charges using AFM-based methods has been reported in a variety of materials,^{24–27} only a few studies have been implemented to probe the interfacial charge trapping in dielectric polymer nanocomposites. Borgani et al. tested low-density polyethylene (LDPE)/Al₂O₃ nanocomposite with a modified EFM (ImEFM) method.¹⁹ They found that the surface potential of the interface is significantly different from the LDPE matrix, and supposed that there were localized states in the interface, that is, the traps. Zhou et al. observed an electric interfacial region around an M-SiO₂@Al₂O₃ particle in epoxy from the result of surface potential using KPFM.²³ They found that the charges preferred to accumulate around the core–shell-structured mesoporous-SiO₂@Al₂O₃ (M-SiO₂@Al₂O₃) single particle and the size of the interfacial region was about 50 nm. Although these experimental results are exciting, it must be noted that the conventional EFM or open-loop KPFM methods cannot distinguish the effect of space charges from that of materials' electrical polarization. This is because the biased probe exerting a vertical direct current (DC) electric field can induce electrical polarization of the sample in the same direction, and the intensity of polarization is usually different between the interfacial region and the polymer matrix,^{20,22} which would affect the gauged electrostatic force. This issue would lead to significant discrepancies or even incorrect data in the detection of local charge behavior. Unfortunately, this issue has been

ignored in previous studies, and precisely probing the interfacial space charge in polymer nanocomposites remains challenging, especially when polar polymers (e.g., ferroelectrics) are involved. Castellon et al. developed a method to quantify space charge from vertical polarization at different lift heights and voltages, which is potentially useful in precise charge mapping.²⁸ However, the approximation in that method requires the probe-sample arrangement to be axisymmetric, which may not be suitable for heterogeneous dielectric materials. In addition, this method involves multiple tests at each point with different voltages, which may consume more time and may find difficulty in time-resolved measurements.

In this study, an open-loop KPFM method is developed to probe the space charge profile at nanoscale in polymer nanocomposites. To conduct the detection of interfacial trap properties and dynamics, a DC electric field needs to be applied on the sample during the KPFM characterization to generate charge injection and to drive charge transport. Instead of applying the DC field via the AFM tip in the conventional KPFM methods,^{22,23} we fabricate a pair of extrinsic electrodes on a silicon substrate to mount the sample and apply the driving electric field in the horizontal direction. This design allows charge carriers to be injected and to migrate in the polymer while inhibiting any electrical polarization in the vertical direction, which enables probing of local traps at the nanoscale without the interference of electrical polarization.

2. RESULTS AND DISCUSSION

2.1. Theory of Open-Loop KPFM. The configuration of electrodes is schematically illustrated in Figure 1a, and the scanning electron microscopy (SEM) images are shown in Figure 1b. The silicon substrate with sufficient flatness is utilized to meet the needs of nanoscale AFM measurement. A dense SiO₂ layer with a thickness of about 2 μm is on top of the silicon substrate as the electrical insulation. A pair of electrodes, with a thickness of about 120 nm, are processed by lithography of metal aluminum that is vapor-deposited on the surface of the SiO₂ layer. To reduce the possibility of short circuit, the aluminum electrodes are designed to be T-shaped, where a slit with length of 50 μm and width of less than 2 μm

is formed between the two electrodes. The solution of the nanocomposite is drop-cast inside the slit using a pipette and then is dried to form a thin film for detection (Figure 1c). The narrow slit is beneficial for applying a sufficiently high lateral electric field between the electrodes without the need for very high voltages that are likely to damage the equipment.

The schematic diagram of the working principle for the lift mode, open-loop KPFM is depicted in Figure 1d. The measurement is performed through a double-pass scan. In the first scan, the topography information of the sample is recorded with the PeakForce tapping mode imaging, which is used in the second scan to retrace the baseline with the probe lifted at a given height above the sample. A switching device is designed to apply the external voltage between the two electrodes or to short them, where the right electrode is grounded by default and the voltage is applied to the left one. Once the external voltage is applied, the lateral electric potential distribution, denoted as $V(x)$, is generated in the film laid inside the slit, where x presents the distance from the grounded electrode. It is apparent that $V(x)$ is linearly distributed under the lateral electric field in an electrically homogeneous material (such as in the blank polymer). The electric field drop at the electrode/polymer interface due to the contact resistance is negligible even with such a short insulation distance (i.e., slit width) according to the result of electric polarization measurement (Figure S1), which should be attributed to the fact that the electrical resistivity of dielectric materials is very high in comparison to the contact resistance at the electrode/polymer interface. At the same time, an AC voltage was applied to the tip of KPFM and the electric potential difference between the tip and sample can be expressed as

$$V_{\text{tip}} = V_{\text{ac}} \sin(\omega t) - \Delta V - V(x) \quad (1)$$

where $V_{\text{ac}} \sin(\omega t)$ is the AC voltage applied to the tip, $V(x)$ is the lateral electric potential between the slit, which is generated by the external voltage applied on the electrodes, and $\Delta V = \phi_{\text{sample}} - \phi_{\text{tip}}$ represents the intrinsic surface potential difference between the tip and the sample, which is related to the difference of work function between the tip and the sample. Since the lateral electric field injects charges and drives charge transport, it is obvious that there is charge distribution inside and on the surface of the sample. As a result, in the vertical direction, the total electrostatic force on the tip comprises two components, which are the capacitive force between the tip and the sample (F_{cap}) and the Coulomb force generated by surface charges of the sample (F_{Coul}), and can be written as²⁹

$$F_{\text{tot}} = F_{\text{cap}} + F_{\text{Coul}} = \frac{1}{2} \frac{\partial C}{\partial z} V_{\text{tip}}^2 + E_z Q_{\text{tip}} \quad (2)$$

where z is the lift height of the tip, C is the capacitance between the probe/cantilever and the substrate, E_z is the vertical component of the electric field generated by the free surface charges of the sample, and Q_{tip} is the sum of the charge on the capacitor CV_{tip} plus the image charges Q_{im} induced by the charge distribution on the surface of the sample.²⁴ Therefore, the total electrostatic force on the tip can be written as

$$\begin{aligned} F_{\text{tot}} &= \frac{1}{2} \frac{\partial C}{\partial z} (V_{\text{ac}} \sin(\omega t) - \Delta V - V(x))^2 \\ &\quad + E_z (Q_{\text{im}} + C(V_{\text{ac}} \sin(\omega t) - \Delta V - V(x))) \\ &= \frac{1}{2} \frac{\partial C}{\partial z} \left((V(x) + \Delta V)^2 + \frac{1}{2} V_{\text{ac}}^2 \right) \\ &\quad + E_z (Q_{\text{im}} - C(V(x) + \Delta V)) \\ &\quad + \left(-\frac{\partial C}{\partial z} (V(x) + \Delta V) + CE_z \right) V_{\text{ac}} \sin(\omega t) \\ &\quad - \frac{1}{4} \frac{\partial C}{\partial z} V_{\text{ac}}^2 \cos(2\omega t) \end{aligned} \quad (3)$$

from which the ω and 2ω components of electrostatic force can be derived

$$F_{\omega} = \left(-\frac{\partial C}{\partial z} (V(x) + \Delta V) + CE_z \right) V_{\text{ac}} \sin(\omega t) \quad (4)$$

$$F_{2\omega} = \frac{1}{4} \frac{\partial C}{\partial z} V_{\text{ac}}^2 \cos(2\omega t) \quad (5)$$

Since the van der Waals force is a kind of short-range force, whereas the electrostatic force is long-ranged, when the probe is lifted, the van der Waals force decreases rapidly, and the interaction between the probe and the sample is mainly the electrostatic force.³⁰ Due to the impact of interaction force, the vibration of the probe can be described as a forced Lorentz oscillator model,³¹ and it can be deduced that (see the Supporting Information for details)

$$A = \frac{|F_d|}{m \sqrt{\left(\frac{\omega_0 \omega}{Q} \right)^2 + (\omega_0^2 - \omega^2)^2}} \quad (6)$$

where Q and ω_0 are the quality factor and resonance angular frequency of the probe cantilever, respectively, and F_d and ω are the amplitude and angular frequency of driving force (electrostatic force) applied on the tip. When ω and 2ω are far less than ω_0 , the denominator in eq 6 can be regarded as constant, and therefore $A \propto |F|$, and we get the following two equations:

$$A_{\omega} \propto |F_{\omega}| = \left| \frac{\partial C}{\partial z} (V(x) + \Delta V) - CE_z \right| V_{\text{ac}} \quad (7)$$

$$A_{2\omega} \propto |F_{2\omega}| = \frac{1}{4} \frac{\partial C}{\partial z} V_{\text{ac}}^2 \quad (8)$$

Equations 4 and 7 show that F_{ω} and A_{ω} are related to the lateral electrical potential $V(x)$ and the contribution of free surface charges CE_z , which means that they are actually influenced by the lateral electric field. And if we can extract the CE_z signal (contribution from the free charge) under the lateral electric field, the mapping of free space charges will be achievable.

2.2. Sample Preparation for the KPFM Study. We testify the effectiveness of this method on highly polar ferroelectric polymer nanocomposites, which are widely used in capacitive energy storage, electrocaloric refrigeration, and mechanical energy harvesting.^{32–36} In the conventional KPFM measurement under the vertically applied DC electric field, the strong electrical polarization of ferroelectric materials would

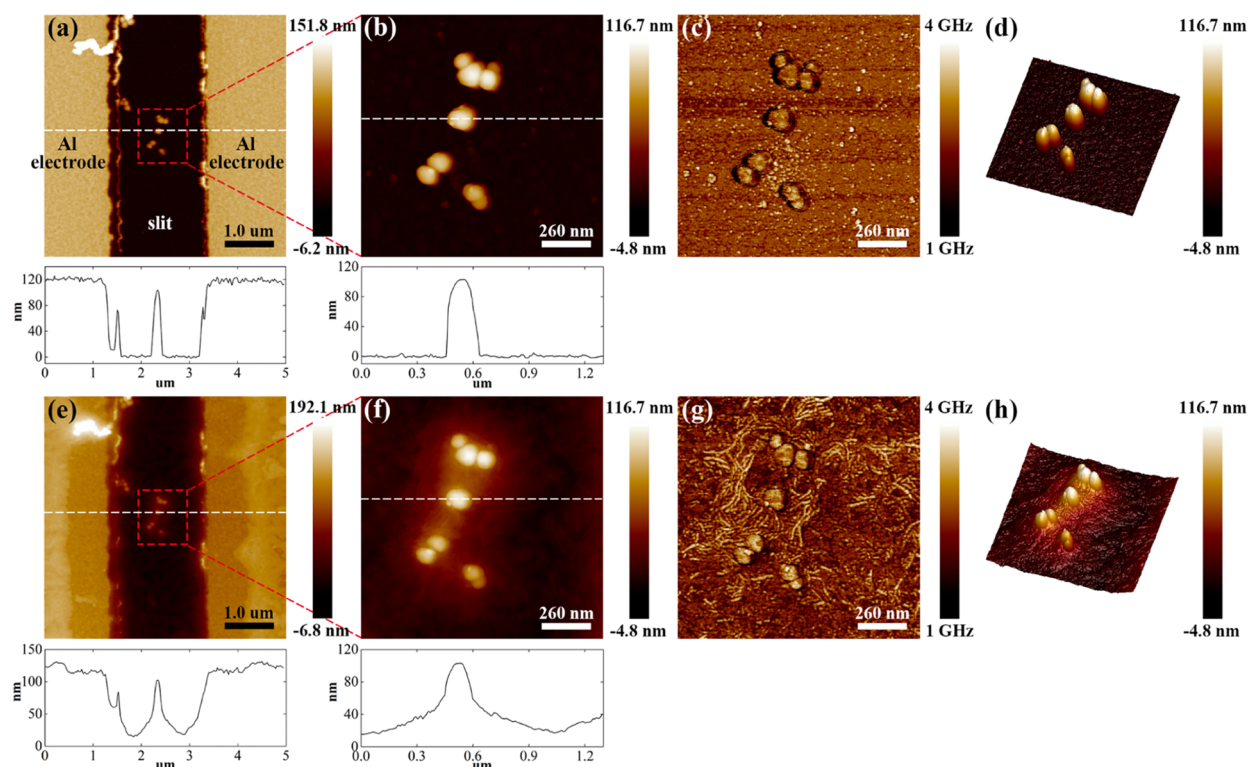


Figure 2. (a, b) Topographies of the nanoparticles inside the slit. (c) Mapping of elastic modulus and (d) three-dimensional topography of the same tested microarea as that in (b). (e, f) Topographies of the nanocomposite film inside the slit. (g) Mapping of elastic modulus and (h) three-dimensional topography of the same tested microarea as that in (f).

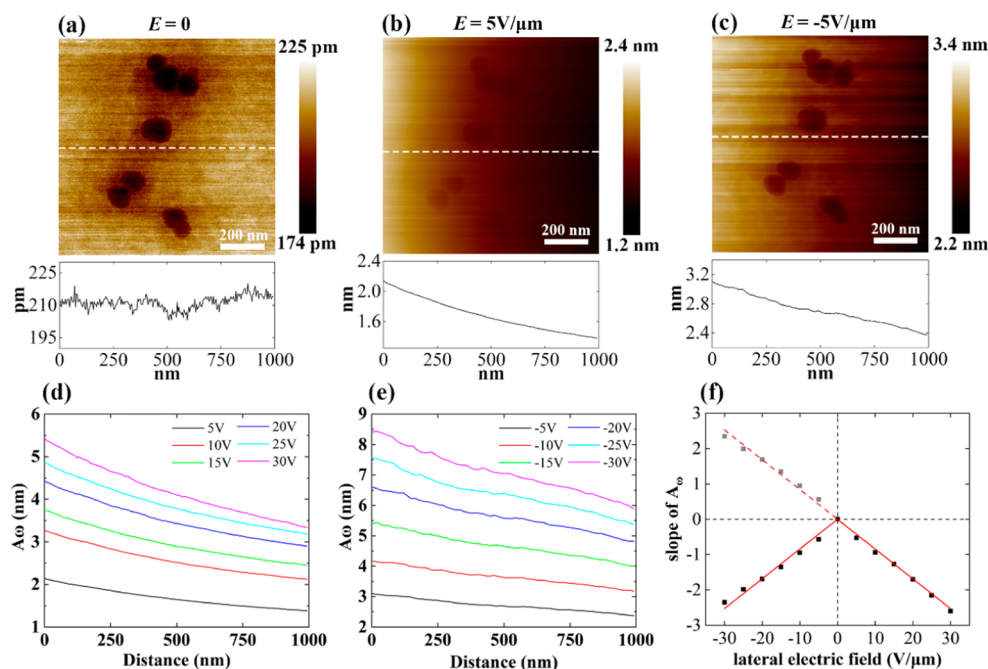


Figure 3. (a) A_ω signal without the applied lateral electric field. (b, c) A_ω signals with 5 and -5 V/μm applied lateral electric field, respectively. (d, e) Distribution of A_ω signals along the horizontal direction under the positive and negative lateral electric fields, respectively. (f) Linear relationship between the slope of A_ω and the applied lateral electric field.

significantly overwhelm the detection of local charge trapping. The copolymer of vinylidene fluoride and trifluoroethylene (P(VDF-TrFE)), and barium titanate (BT) nanoparticles are chosen as the polymer matrix and nanofillers, respectively. The sample for KPFM study was prepared in two steps. In the first

step, the *N,N*-dimethylformamide (DMF) suspension of BT nanoparticles was dropped between the two electrodes using a pipette and dried, which was then observed using AFM to ensure that BT nanoparticles were deposited within the slit. The typical topography with satisfactory sample preparation is

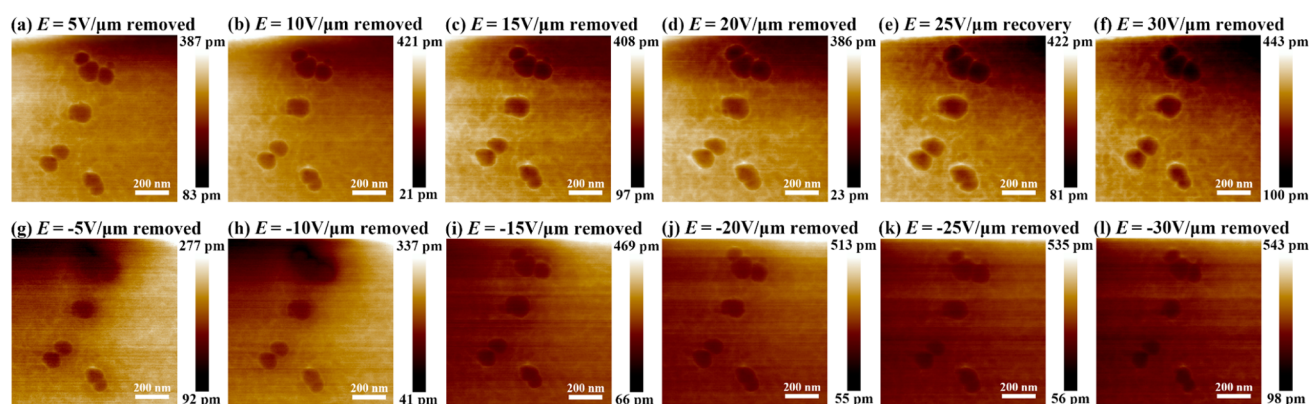


Figure 4. A_{ω} signals right after the removal of the (a–f) positive and (g–l) negative lateral electric fields. The A_{ω} signals change up and down with increasing lateral electric field because the switch-off of the lateral electric field takes a short period of time (approximately several seconds) which cannot be precisely controlled.

shown in Figure 2a–d. It is clear that there are a few nanoparticles right in between the slit. The height of the Al electrode is about 120 nm, and the diameters of BT nanoparticles are about 100 nm. In the second step, we filled the slit with the P(VDF-TrFE) polymer. To expose the interfacial region for detection, we drop-cast with a pipette the ultradiluted DMF solution of P(VDF-TrFE) of precisely controlled concentration and dried it in vacuum. The topography and mapping of elastic modulus of a typical sample are shown in Figure 2e–h. It is not difficult to see through the crystalline nanofibril structures and the height increment surrounding the nanoparticles that the P(VDF-TrFE) polymer was successfully coated on the substrate in the slit. The average thickness of the polymer film is about 30 nm, which does not exceed the diameter of the BT nanoparticles and thus exposes the interfacial regions.

2.3. Interpretation of the Output Signal. To establish the mapping of free space charges, we carefully studied the output signal of KPFM with changing applied lateral electric field (from positive to negative with a 5 V/μm interval). We first noted that the $A_{2\omega}$ signal remained constant with changing lateral electric field (Figure S2). This result is consistent with eqs 5 and 8 that both $F_{2\omega}$ and $A_{2\omega}$ are only related to the AC voltage applied to the tip and are not affected by the lateral electric potential $V(x)$ or the free surface charges of the sample CE_z . Therefore, the fact that $A_{2\omega}$ remained unchanged with changing lateral electric field confirms the reliability of the signal analytical method. Because the measured $A_{2\omega}$ signals are quite uniform in the different regions within the polymer matrix, it is also conclusive that $\partial C/\partial z$ is hardly affected by the minor surface fluctuation of the polymer matrix.

We then focused on the A_{ω} signal. According to eq 7, $A_{\omega}(x)$ is determined by four influencing factors under the applied lateral field, i.e., $\partial C/\partial z$, $V(x)$, ΔV , and CE_z . Our purpose is to understand the CE_z signal (contribution from the free charge) under the lateral electric field. The results of A_{ω} with and without applied lateral electric field are depicted in Figures 3a–c and S3. According to eq 7, without the lateral electric field and under constant AC amplitude, A_{ω} is only related to $\partial C/\partial z$ and ΔV ($A_{\omega} \propto (\partial C/\partial z)|\Delta V|V_{ac}$). The $\partial C/\partial z$ signal in the polymer matrix has been proved to be not affected by the topography (due to the minor surface fluctuation) or the lateral electric field (Supporting Information Note 2 and Figure S2). The ΔV signal can be considered to be only related to the work function difference ($\Delta\Phi$) between the sample and

the tip, and should be a constant value in the polymer matrix and interface (Supporting Information Note 3). It is found that without the applied lateral electric field, the measured A_{ω} signals of the matrix and interface are basically the same (Figure 3a), suggesting that $\partial C/\partial z$ and ΔV are indeed constant values in the polymer matrix and interface. Note that A_{ω} is experimentally found to be directly proportional to the AC amplitude, whereas $A_{2\omega}$ shows a quadratic correlation with the AC amplitude (Figures S4–S6), which comply very well with eqs 7 and 8, respectively. These results suggest that AC bias on the probe does not affect the free surface charge and thus CE_z .

According to the aforementioned results, under the applied lateral DC electric field and constant AC amplitude on the tip, only $V(x)$ and CE_z signals are variables in the analysis of A_{ω} . As seen from the KPFM result, when the lateral electric field is applied, the A_{ω} signal exhibits a gradual reduction from the high-voltage electrode to the ground one (Figure 3b–e), and therefore A_{ω} can be regarded as a function of x (the distance from the grounded electrode), denoted as $A_{\omega}(x)$. In a homogeneous material (such as in the pure polymer), $V(x)$ should, in principle, be linearly distributed under the applied lateral electric field, and the slope of $V(x)-x$ should be proportional to the strength of the external lateral electric field. If we scan over the nanoparticles, then there will be fluctuations in $V(x)$ (Figure S7). On the other hand, it is seen from the KPFM result that the slope of $A_{\omega}(x)-x$ is proportional to the strength of lateral electric field as well, as plotted in Figure 3f. This result suggests that $V(x)$ has a major influence on the A_{ω} result, which overshadows the contribution from CE_z , i.e., the charge distribution.

With all of the above results in mind, we next need to separate the contribution of CE_z from that of $V(x)$, to obtain the charge distribution profile. To this end, we investigated the A_{ω} signals as the external voltage is removed, where $V(x)$ would disappear immediately upon the removal of voltage, whereas the free surface charges would dissipate much more slowly, and the A_{ω} signals were only affected by CE_z ($A_{\omega} \propto |\partial C/\partial z|\Delta V - CE_z|V_{ac}|$, where $\partial C/\partial z$ and ΔV are constant values in the polymer matrix). The results of A_{ω} right after the removal of the lateral electric field are depicted in Figure 4. Evidently, A_{ω} changes in both the horizontal and longitudinal directions. The change in the horizontal direction is very similar to that of A_{ω} under the applied lateral electric field, indicating that there is a spatial distribution of A_{ω} , while the change in the longitudinal direction indicates the change of A_{ω}

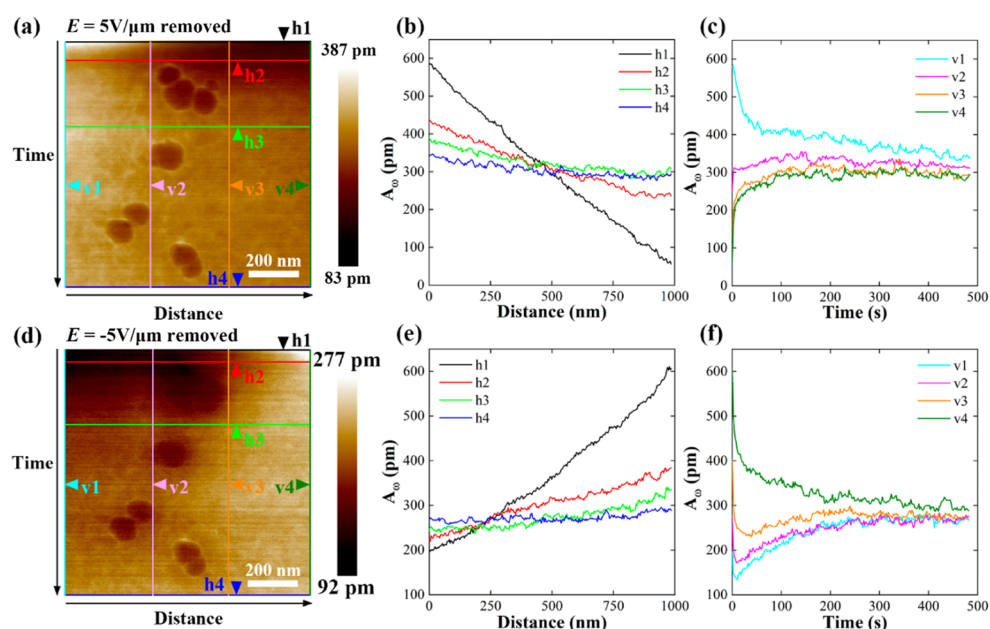


Figure 5. (a) A_{ω} signal right after the removal of the $5 \text{ V}/\mu\text{m}$ lateral electric field and its (b) spatial distribution and (c) change with time. (d) The A_{ω} signal right after the removal of the $-5 \text{ V}/\mu\text{m}$ lateral electric field and its (e) spatial distribution and (f) change with time.

with time. In the KPFM detection, the entire microarea ($1 \mu\text{m} \times 1 \mu\text{m}$) is divided into 256 scan lines in the longitudinal direction, and the probe is scanned line by line from top to bottom. The time interval between each scan line is about 1.9 s, which means that in the KPFM results, there is a time difference of 1.9 s between each line in the longitudinal direction. Note that the A_{ω} signals under positive voltage (Figure 3b) are lower than those under negative voltage (Figure 3c). According to eq 7, we can obtain that ΔV is negative. Then, comparing Figure 4 and Figure 3a, we can find that positive charges accumulated in the area with A_{ω} smaller than $\sim 210 \text{ pm}$ and negative charges accumulated in the area with A_{ω} larger than $\sim 210 \text{ pm}$. It suggests that, roughly, negative charges exist in the light area and positive charges are located in the dark area. Based on the results shown in Figures 3 and 4, we attribute the charges accumulated in the left-hand side of Figure 4a–f and those accumulated in the right-hand side of Figure 4g–i to either negative ions or injected/excited electrons trapped near the electrode (although the applied electric fields in this study are relatively low, there will still be a small number of injected/excited charge carriers).

To further analyze the spatial distribution and time-variant characteristic of A_{ω} after the removal of the lateral electric field, we used the results from removing the $\pm 5 \text{ V}/\mu\text{m}$ lateral field (Figure 4a,g) and selected four horizontal lines (h1, h2, h3, and h4) and four longitudinal lines (v1, v2, v3, and v4) in the images (Figure 5a,d). The selection of lines should avoid the nanoparticles to understand the signal change in the polymer matrix. The corresponding data are summarized in Figure 5b,c,e,f. It is not difficult to see from Figure 5b,e that after removing the lateral electric field, the A_{ω} signals are unevenly distributed in the horizontal direction, while the difference in slopes of the four horizontal lines indicates that the A_{ω} signals also change with time. On the other hand, Figure 5c,f reveals that the A_{ω} signals decay quickly. About 200 s after removing the lateral electric field, the decay rates gradually stabilize and the A_{ω} signals approach the original value before the application of lateral electric field. The time-variant character-

istic of A_{ω} is very similar to the relaxation characteristic of the surface charge: once the external voltage is removed, the distribution balance of surface charges is broken, the charges begin to dissipate through the aluminum electrode, and the rate of dissipation gradually decreases with time and finally reaches equilibrium. The variation trend of A_{ω} shown in Figure 5c,f can be attributed to the different mobility and concentration gradient between electrons and impurity ions that lead to the time-dependent properties. In the case of positive applied voltages shown in Figure 5a–c, the injection of electrons is minor from the ground side (the main intrinsic carrier for P(VDF-TrFE) should be electrons; thus, the effect of holes is ignored here) and the charge behavior is dominated by impurity ions. Negative ions accumulate near the high-voltage side (near the line v1), and positive ions accumulate near the ground side (near the line v4). Then, after the voltage is withdrawn and electrodes are short-circuited, negative ions diffuse as driven by the established concentration gradient from the v4 to v1 side, and similarly, positive ions diffuse from the v1 to v4 side. Thus, in Figure 5c, A_{ω} at v1 decreases (negative carriers decrease) and those at v2, v3, and v4 increase (negative carriers increase). In the case of negative applied voltages shown in Figure 5d–f, the charge behavior is determined by both ions and injected electrons. First for impurity ions, opposite to the positive case, positive ions accumulate near the v1 side and negative ions accumulate near the v4 side. Then for electrons, they are injected from the high-voltage side and trapped near the v1 side. Then, after the negative voltage is withdrawn, both electrons and ions diffuse as driven by the respective concentration gradient. It has been proved that the relaxation time of electrons is much shorter and the mobility is much higher than that of ions. Thus, as electrons quickly diffuse from the v1 to v4 side, in Figure 5f, the A_{ω} values at v1, v2, and v3 show a very rapid decrease in the initial several seconds (negative carriers decrease). As time elapses, the effect of ion diffusion begins to dominate, i.e., positive ions diffuse from the v1 to v4 side and negative ions diffuse the v4 to v1 side. Thus, after a rapid decrease, the A_{ω}

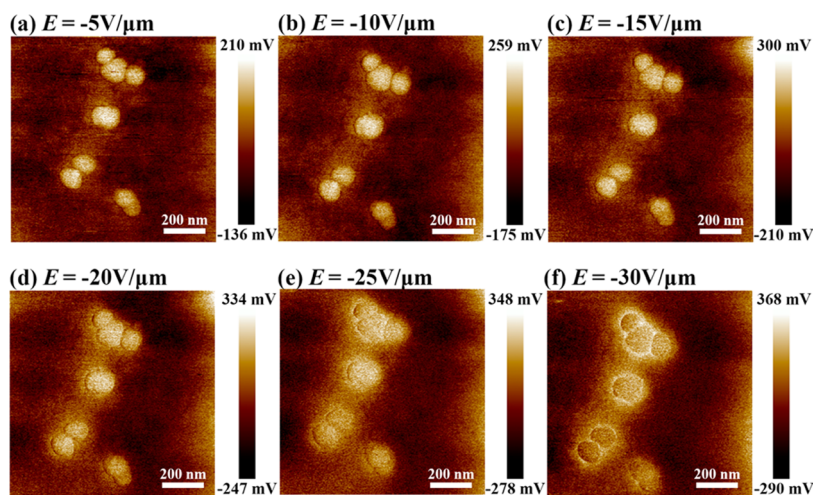


Figure 6. V_{flatten} signals under the negative applied lateral electric fields of (a) $-5 \text{ V}/\mu\text{m}$, (b) $-10 \text{ V}/\mu\text{m}$, (c) $-15 \text{ V}/\mu\text{m}$, (d) $-20 \text{ V}/\mu\text{m}$, (e) $-25 \text{ V}/\mu\text{m}$, and (f) $-30 \text{ V}/\mu\text{m}$ corresponding to the space charge distribution.

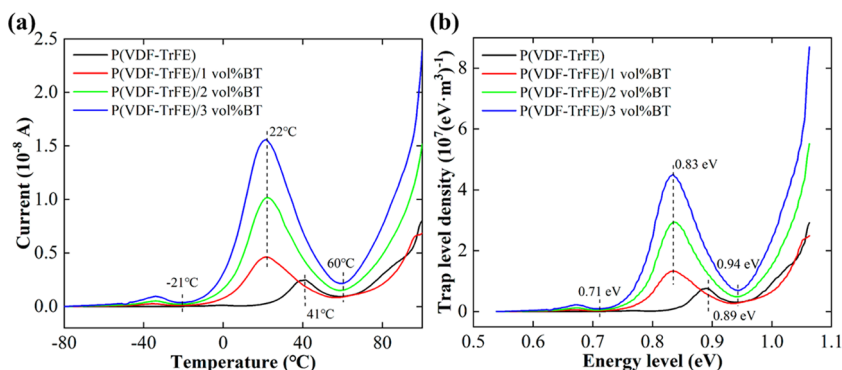


Figure 7. (a) Temperature–current spectra and (b) the trap energy-level density spectra of P(VDF-TrFE) and its nanocomposites.

values at v1, v2, and v3 increase (positive carriers decrease) and meanwhile A_{ω} at v4 decreases.

The line h1 in Figure 5a,d corresponds to the time mark almost immediately after removing the lateral electric field (since the scanning needs to be started after the electric field is removed, the actual time difference is about a few seconds). The A_{ω} data corresponding to the line h1 are unevenly distributed in the horizontal direction at the moment when lateral electric field is removed (Figure 5b,e). Due to the relaxation characteristic of free surface charges, the distribution cannot change abruptly, and therefore it can be inferred that the data of line h1 should be very similar to the equilibrium distribution of free surface charges when the lateral field is still on. Therefore, if we can subtract the regular base signals (the linear component) in the polymer matrix caused by the lateral electric field, then we will be able to extract the irregular space charge accumulation (trapped charges) in the nanocomposites. To this end, we first use eqs 7 and 8 to construct a potential signal, denoted as V_t to reduce the influence of topography, as shown in eq 9, which is a common data processing method in KPFM detection

$$V_t = \frac{1}{4} V_{ac} \frac{A_{\omega}}{A_{2\omega}} = \frac{1}{4} V_{ac} \frac{|E_{\omega}|}{|E_{2\omega}|} = \left| V(x) + \Delta V - \frac{CE_z(x)}{\partial C / \partial z} \right| \quad (9)$$

And then, we can simplify the expressions for V_t into

$$V_t = \left| V(x) + \Delta V - \frac{CE_z(x)}{\partial C / \partial z} \right| = |a_1 + b_1x + \Delta V - (a_2 + b_2x + c_2(x))| = |(a_1 - a_2 + \Delta V) + (b_1 - b_2)x - c_2(x)| \quad (10)$$

where a_1 , a_2 , b_1 , and b_2 are all constants independent of x , $a_1 + b_1x$ represents the linear nature of $V(x)$, $a_2 + b_2x$ stands for the linear component of CE_z , and $c_2(x)$ corresponds to the nonlinear component of CE_z (and hence the irregular space charge accumulation). If we perform a linear fit to V_t and subtract the result of the fitting (the process is denoted as “flatten”), i.e., to subtract the linear part of the base signals, the obtained signal (noted as V_{flatten}) will mainly reflect the trapped space charge

$$V_{\text{flatten}} = V_t - |(a_1 - a_2 + \Delta V) + (b_1 - b_2)x| = \pm c_2(x) \quad (11)$$

In such signal processing to the polymer nanocomposites, we can reveal the charge profile where the charge distribution is modified by the heterogeneous structures, e.g., the trapping sites in the interfacial region. We perform the above-mentioned signal processing in each test to obtain the V_{flatten} signal, and the results are shown in Figures 6 and S8. Interestingly, when a sufficiently high negative lateral electric field ($>20 \text{ V}/\mu\text{m}$) is applied, as shown in Figure 6e,f, significant halo signals with the range of about 20–30 nm are observed around the

nanoparticles. Clearly, since no vertical polarization is induced in the test, these irregular signals can be ascribed to the space charge trapping in the interfacial region of the polymer nanocomposite (see Supporting Information Note 4 for more details).

2.4. Verification of the Charge Mapping. We finally employed one of the common analytical techniques for characterizing charge trapping, i.e., TSDC to examine the reliability of the KPFM method. The P(VDF-TrFE)/BT nanocomposite film used for TSDC test was prepared by spin coating, with a thickness of about 50 μm . And the trap energy-level density spectra are recorded using the obtained temperature–current curves,³⁷ as shown in Figure 7. When the temperature approaches 100 $^{\circ}\text{C}$, the current increases rapidly in all of the samples, which is attributed to the ferroelectric–paraelectric phase transition of P(VDF-TrFE). It can be seen that the P(VDF-TrFE) matrix has a signal peak at 41 $^{\circ}\text{C}$, which corresponds to the depolarization of the injected charges trapped in the amorphous–crystalline interface.³⁸ After adding the BT nanoparticles, the nanocomposites have new, strong signal peaks centered at 22 $^{\circ}\text{C}$, and the current and trap density of the peak increase almost proportionally with the increase of the content of BT nanoparticles. These new peaks are most likely due to the depolarization of charges trapped in the matrix–particle interfacial region. Upon the addition of the nanoparticles, free charges are largely captured by these matrix–particle interfacial traps such that the original signal peak of the polymer at 41 $^{\circ}\text{C}$ is weakened. The results of the TSDC show that the energy level of the matrix–particle interfacial traps ranges from 0.71 to 0.94 eV, and the corresponding temperature range is from -21 to 60 $^{\circ}\text{C}$. This means that at room temperature (about 25 $^{\circ}\text{C}$), still a large number of charges are being trapped, resulting in the strong signal in the interfacial region in the KPFM detection. On the other hand, these trapped charges will gain sufficient energy to escape from the interfacial traps if the temperature is at or above 60 $^{\circ}\text{C}$. To verify this point, the KPFM test was implemented again at 60 $^{\circ}\text{C}$. To avoid deformation of the sample caused by uneven heating, the entire substrate was heated by an external electric heater, and the probe was heated by the internal temperature control module. The relevant parameters were adjusted to keep the sample at a constant temperature of 60 $^{\circ}\text{C}$. The detection used the same parameters as those at room temperature, and the V_{flatten} signals under the applied lateral electric field were obtained using the same signal processing method. It can be seen clearly from the results that the halo signals generated by the trapped charges in the interfacial region disappear (Figure S9), which proves that under the temperature of 60 $^{\circ}\text{C}$, the interfacial traps can no longer effectively capture the charges. This is consistent with the results of the TSDC test and verifies that the mapping signals recorded by the present KPFM method indeed come from the trapped charges.

3. CONCLUSIONS

This work introduced an open-loop KPFM method for probing the space charge at nanoscale in polymer nanocomposites. By employing a delicate electrode configuration, a lateral electric field was allowed to be applied on the sample during the KPFM test, enabling real-time space charge injection and transport. Since no vertical DC electric field was applied on the AFM probe, the interference from induced electrical polarization was eliminated. Capitalizing on the

KPFM method, we successfully obtained the space charge profile in the polymer nanocomposites. The experimental results show that the space charges are trapped around the embedded nanoparticles, forming a halo charge layer with a thickness of 20–30 nm. The dynamic process by which the trapped charges gradually dissipate after the removal of the lateral electric field was also recorded. TSDC results revealed that the detrapping of the interfacial charges in the polymer nanocomposites occurred at 60 $^{\circ}\text{C}$, which is consistent with the results of the KPFM test, corroborating the reliability of the KPFM method in the detection of space charge. This work provides a robust method for studying the space charge distribution and dynamics at nanoscale resolution, which opens up new possibilities in understanding the various interfacial regions in heterogeneous dielectric materials.

4. EXPERIMENTAL METHODS

The sample for KPFM study was prepared in two steps. In the first step, 1 μL of the DMF (99.9% MACKLIN) suspension (0.1 mg/mL) of BT nanoparticles (100 nm diameter in average, Aladdin Industrial Corporation) was dropped between the two electrodes using a pipette and then the whole substrate was baked at 60 $^{\circ}\text{C}$ in a vacuum oven for 30 min. In the second step, 1 μL of the undiluted DMF solution (0.1 mg/mL) of P(VDF-TrFE) (70:30 VDF/TrFE molar ratio, PIEZOTECH) was dropped between the two electrodes with a pipette. Finally, the whole substrate was again baked at 60 $^{\circ}\text{C}$ in a vacuum oven to ensure that the solvent was completely removed.

Electric polarization–electric field loops were measured at 10 Hz by a ferroelectric test system (PolyK Technologies).

The sample for TSDC measurement was prepared by spin coating a DMF solution of the polymer or the nanocomposite at 500 rpm, with the resultant film thickness of about 50 μm .

TSDC was measured using a Keithley 6517B electrometer as follows. The sample is polarized at room temperature (25 $^{\circ}\text{C}$) with a polarization voltage of 20 kV/mm; then, the sample is quickly cooled to -80 $^{\circ}\text{C}$ and short-circuited for 10 min to eliminate the influence of stray charges. After that, the sample is heated to 100 $^{\circ}\text{C}$ at a rate of 3 $^{\circ}\text{C}/\text{min}$, and the depolarization current is measured at the same time, with a temperature resolution of 1 $^{\circ}\text{C}$.

The probe used in the test is PFQNE-AL made of doped silicon with a high electrical conductivity. The external voltage for generating the lateral electric field was applied by a Keithley 6517B electrometer. The parameters used in the AFM detection are shown in Table 1.

Table 1. Experimental Parameters for the Trap Property of the Interface

parameters	values
tip AC voltage (V_{ac})	4 V
tip AC frequency (f)	10 kHz
resonance frequency of the probe (f_0)	about 300 kHz
lift height of the probe (z)	50 nm
lateral electric field (E)	0 (short circuit), ± 5 , ± 10 , ± 15 , ± 20 , ± 30 V/ μm

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.0c13669>.

Additional notes discussing the forced Lorentz oscillator model, the $A_{2\omega}$ signal, the surface potential difference signal ΔV , and the local CE_z signal; characterizations of $A_{2\omega}$ signals with and without lateral electric field; A_{ω} signals with the applied lateral electric field; V_{flatten}

signals under the positive lateral electric field; and V_{flatten} signals under lateral electric field at 60 °C (PDF)

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Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Kao, K. C.; Hwang, W. *Electrical Transport in Solids*; Pergamon: Oxford, 1981.
- (2) Roy, M.; Nelson, J. K.; MacCrone, R. K.; Schadler, L. S.; Reed, C. W.; Keefe, R. Polymer Nanocomposite Dielectrics—the Role of the Interface. *IEEE Trans. Dielectr. Electr. Insul.* **2005**, *12*, 629–643.
- (3) Luo, H.; Zhou, X.; Ellingford, C.; Zhang, Y.; Chen, S.; Zhou, K.; Zhang, D.; Bowen, C. R.; Wan, C. Interface Design for High Energy Density Polymer Nanocomposites. *Chem. Soc. Rev.* **2019**, *48*, 4424–4465.
- (4) Tanaka, T.; Kozako, M.; Fuse, N.; Ohki, Y. Proposal of a Multi-core Model for Polymer Nanocomposite Dielectrics. *IEEE Trans. Dielectr. Electr. Insul.* **2005**, *12*, 669–681.
- (5) Lewis, T. J. Interfaces are the Dominant Feature of Dielectrics at the Nanometric Level. *IEEE Trans. Dielectr. Electr. Insul.* **2004**, *11*, 739–753.
- (6) Pourrahimi, A. M.; Hoang, T. A.; Liu, D.; Pallon, L. K. H.; Gubanski, S.; Olsson, R. T.; Gedde, U. W.; Hedenqvist, M. S. Highly Efficient Interfaces in Nanocomposites Based on Polyethylene and ZnO Nano/Hierarchical Particles: A Novel Approach toward Ultralow Electrical Conductivity Insulations. *Adv. Mater.* **2016**, *28*, 8651–8657.
- (7) Meunier, M.; Quirke, N. Molecular Modeling of Electron Trapping in Polymer Insulators. *J. Chem. Phys.* **2000**, *113*, 369–376.
- (8) Meunier, M.; Quirke, N.; Aslanides, A. Molecular Modeling of Electron Traps in Polymer Insulators: Chemical Defects and Impurities. *J. Chem. Phys.* **2001**, *115*, 2876–2881.
- (9) Brodribb, J. D.; O'Colmain, D.; Hughes, D. M. The Theory of Photon-Stimulated Current Spectroscopy. *J. Phys. D: Appl. Phys.* **1975**, *8*, 856–862.
- (10) McKeever, S. W.; Lilley, E. Thermally Stimulated Polarisation and Depolarisation Currents (TSPC and TSDC) in LiF:Mg²⁺ (TLD-100). *J. Phys. C: Solid State Phys.* **1981**, *14*, 3547–3555.
- (11) Simmons, J. G.; Tam, M. C. Theory of Isothermal Currents and the Direct Determination of Trap Parameters in Semiconductors and Insulators Containing Arbitrary Trap Distributions. *Phys. Rev. B* **1973**, *7*, No. 3706.
- (12) Nalbach, M.; Kliem, H. Contact Charging and Surface Charge Measurement Using a Scanning Kelvin Technique. *Phys. Status Solidi A* **2000**, *178*, 715–719.
- (13) Cheran, L. E.; Lam, P.; Zheng, Z.; Boggs, S.; Thompson, M. In *Characterization of Dielectric Materials Using a High-Resolution Scanning Kelvin-Microprobe*, Annual Report of the IEEE Conference on Electrical Insulation and Dielectric Phenomena; 2003; pp 16–19.
- (14) Mizutani, T. Space Charge Measurement Techniques and Space Charge in Polyethylene. *IEEE Trans. Dielectr. Electr. Insul.* **1994**, *1*, 923–933.
- (15) Takada, T. Acoustic and Optical Methods for Measuring Electric Charge Distributions in Dielectrics. *IEEE Trans. Dielectr. Electr. Insul.* **1999**, *6*, 519–547.
- (16) Bert, V. *Scanning Probe Microscopy*; Springer: New York, 2015.
- (17) Tanimoto, M.; Vatel, O. Kelvin Probe Force Microscopy for Characterization of Semiconductor Devices and Processes. *J. Vac. Sci. Technol., B* **1996**, *14*, 1547–1551.
- (18) Cadena, M. J.; Sung, S. H.; Boudouris, B. W.; Reifenger, R.; Raman, A. Nanoscale Mapping of Dielectric Properties of Nanomaterials from Kiloherztz to Megahertz Using Ultrasmall Cantilevers. *ACS Nano* **2016**, *10*, 4062–4071.
- (19) Borgani, R.; Pallon, L. K.; Hedenqvist, M. S.; Gedde, U. W.; Haviland, D. B. Local Charge Injection and Extraction on Surface-Modified Al₂O₃ Nanoparticles in LDPE. *Nano Lett.* **2016**, *16*, 5934–5937.
- (20) Peng, S.; Zeng, Q.; Yang, X.; Hu, J.; Qiu, X.; He, J. Local Dielectric Property Detection of the Interface between Nanoparticle and Polymer in Nanocomposite Dielectrics. *Sci. Rep.* **2016**, *6*, No. 38978.
- (21) Yang, Y.; Zhang, X.; Qin, L.; Zeng, Q.; Qiu, X.; Huang, R. Probing Nanoscale Oxygen Ion Motion in Memristive Systems. *Nat. Commun.* **2017**, *8*, No. 15173.
- (22) Peng, S.; Yang, X.; Yang, Y.; Wang, S.; Zhou, Y.; Hu, J.; Li, Q.; He, J. Direct Detection of Local Electric Polarization in the Interfacial Region in Ferroelectric Polymer Nanocomposites. *Adv. Mater.* **2019**, *31*, No. 1807722.
- (23) Zhou, J.; Li, Y.; Wu, Y.; Jia, B.; Zhu, L.; Jiang, Y.; Li, Z.; Wu, K. Tuned Local Surface Potential of Epoxy Resin Composites by Inorganic Core-Shell Microspheres: Key Roles of the Interface. *Langmuir* **2019**, *35*, 12053–12060.
- (24) Collins, S.; Mikhnenko, O.; Nguyen, T.; Rengert, Z.; Bazan, G.; Woo, H.; Nguyen, T. Observing Ion Motion in Conjugated

Polyelectrolytes with Kelvin Probe Force Microscopy. *Adv. Electron. Mater.* **2017**, *3*, No. 1700005.

(25) Zhang, Y.; Ziegler, D.; Salmeron, M. Charge trapping states at the SiO₂-Oligothiophene Monolayer Interface in Field Effect Transistors Studied by Kelvin Probe Force Microscopy. *ACS Nano* **2013**, *7*, 8258–8265.

(26) Collins, L.; Vasudevan, R.; Sehirlioglu, A. Visualizing Charge Transport and Nanoscale Electrochemistry by Hyperspectral Kelvin Probe Force Microscopy. *ACS Appl. Mater. Interfaces*. **2020**, *12*, 33361–33369.

(27) Weber, S.; Hermes, I.; Turrencruz, S.; Gort, C.; Bergmann, V.; Gilson, L.; Hagfeldt, A.; Graetzel, M.; Tress, W.; Berger, R. How the Formation of Interfacial Charge Causes Hysteresis in Perovskite Solar Cells. *Energy Environ. Sci.* **2018**, *11*, 2404–2413.

(28) El Khoury, D.; Arinero, R.; Laurentie, J.; Castellon, J. Nanoscale Surface Charge Detection in Epoxy Resin Materials Using Electrostatic Force Spectroscopy. *AIP Adv.* **2016**, *6*, No. 035318.

(29) Cherniavskaya, O.; Chen, L.; Weng, V.; Yuditsky, L.; Brus, L. E. Quantitative Noncontact Electrostatic Force Imaging of Nanocrystal Polarizability. *J. Phys. Chem. B* **2003**, *107*, 1525–1531.

(30) Saint Jean, M.; Hudlet, S.; Guthmann, C.; Berger, J. Van der Waals and Capacitive Forces in Atomic Force Microscopies. *J. Appl. Phys.* **1999**, *86*, 5245–5248.

(31) Garcia, R.; Perez, R. Dynamic Atomic Force Microscopy Methods. *Surf. Sci. Rep.* **2002**, *47*, 197–301.

(32) Li, Q.; Wang, Q. Ferroelectric Polymers and Their Energy-Related Applications. *Macromol. Chem. Phys.* **2016**, *217*, 1228–1244.

(33) Dang, Z.; Yuan, J.; Yao, S.; Liao, R. Flexible Nanodielectric Materials with High Permittivity for Power Energy Storage. *Adv. Mater.* **2013**, *25*, 6334–6365.

(34) Yao, Z.; Song, Z.; Hao, H.; Yu, Z.; Cao, M.; Zhang, S.; Lanagan, M. T.; Liu, H. Homogeneous/Inhomogeneous-Structured Dielectrics and their Energy-Storage Performances. *Adv. Mater.* **2017**, *29*, No. 1601727.

(35) Li, Q.; Zhang, G.; Zhang, X.; Jiang, S.; Zeng, Y.; Wang, Q. Relaxor Ferroelectric-Based Electrocaloric Polymer Nanocomposites with a Broad Operating Temperature Range and High Cooling Energy. *Adv. Mater.* **2015**, *27*, 2236–2241.

(36) Singh, D.; Choudhary, A.; Garg, A. Flexible and Robust Piezoelectric Polymer Nanocomposites Based Energy Harvesters. *ACS Appl. Mater. Interfaces* **2018**, *10*, 2793–2800.

(37) Tian, F.; Bu, W.; Shi, L.; Yang, C.; Wang, Y.; Lei, Q. Theory of Modified Thermally Stimulated Current and Direct Determination of Trap Level Distribution. *J. Electrostat.* **2011**, *69*, 7–10.

(38) Venkatesan, T. R.; Gulyakova, A. A.; Frübing, P.; Gerhard, R. Electrical Polarization Phenomena, Dielectric Relaxations and Structural Transitions in a Relaxor-Ferroelectric Terpolymer Investigated with Electrical Probing Techniques. *Mater. Res. Express* **2019**, *6*, No. 125301.